

Proof of Theorem 2 (continued) .

$$\mathcal{F} = \{ f : [x_0, x_N] \rightarrow \mathbb{R} \mid f \text{ is continuous, } f(x_0) = F_0, f(x_N) = F_N \}$$

$$d(f, g) = \max \{ |f(x) - g(x)| : x \in [x_0, x_N] \}$$

for $f, g \in \mathcal{F}$.

$(\mathcal{F}, d)$ is a complete metric space.

Define $T : \mathcal{F} \rightarrow \mathcal{F}$ by

$$(Tf)(x) = c_n l_n^{-1}(x) + d_n f(l_n^{-1}(x)) + f_n$$

for $x \in [x_{n-1}, x_n], n=1, 2, \dots, N$,

where $l_n(x) = a_n x + c_n$.

$$(l_n : [x_0, x_N] \rightarrow [x_{n-1}, x_n])$$

$$\begin{aligned} Tf(x_n) &= c_n l_n^{-1}(x_n) + d_n f(l_n^{-1}(x_n)) + f_n \\ &= c_n x_N + d_n f(x_N) + f_n \\ &= c_n x_N + d_n F_N + f_n = F_n \end{aligned}$$

for $n = 0, 1, 2, \dots, N$.

Then, Tf interpolates the data set $\{(x_i, F_i) : i = 0, 1, \dots, N\}$.

Now, if $f, g \in \mathcal{F}$,

$$\begin{aligned} d(Tf, Tg) &= \max \{ |Tf(x) - Tg(x)| : x \in [x_0, x_N] \} \\ &= \max_{1 \leq n \leq N} \max \{ |Tf(x) - Tg(x)| : x \in [x_{n-1}, x_n] \} \\ &= \max_{1 \leq n \leq N} |d_n| \max_{x \in [x_{n-1}, x_n]} |f(l_n^{-1}(x)) - g(l_n^{-1}(x))| \end{aligned}$$

$$\therefore d(Tf, Tg) \leq \underbrace{\max_{1 \leq n \leq N} |d_n|}_{< 1} d(f, g)$$

$\Rightarrow T$ is a contraction map on (F, d) .

$\therefore \exists$ a unique fixed point f of T .

Then $f(n) = Tf(n) \quad \forall n \in [x_0, x_N]$

Since $Tf(n) = f_n \quad \forall n = 0, 1, \dots, N$,

$f(n) = f_n \quad \forall n = 0, 1, \dots, N$.

Let $\tilde{\gamma}$ denote the graph of f .

Claim: $\tilde{\gamma} = \bigcup_{n=1}^N \omega_n(\tilde{\gamma})$.

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$$\omega_n \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_n & 0 \\ c_n & d_n \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_n \\ f_n \end{pmatrix}$$

Let $(x, f(n)) \in \tilde{\gamma}$ for some $x \in [x_0, x_N]$

Then $\omega_n(x, f(n)) = (a_n x + e_n, c_n x + d_n f(n) + f_n)$

Now, $f(a_n x + e_n) = Tf(a_n x + e_n)$

$= Tf(f(n))$

$= c_n x + d_n f(n) + f_n$

$\Rightarrow \omega_n(x, f(n)) \in \tilde{\gamma} \quad \forall n \in [x_0, x_N]$.

$\Rightarrow \bigcup_{n=1}^N \omega_n(\tilde{\gamma}) \subseteq \tilde{\gamma}$

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Conversely, if $(x, f(x)) \in \tilde{Q}$, then
 $x \in [x_{n-1}, x_n]$ for some $n=1, 2, \dots, N$.

$$\omega_n(\bar{L}_n(x), f(\bar{L}_n(x)))$$

$$= (a_n \bar{L}_n(x) + c_n, c_n \bar{L}_n(x) + d_n f(\bar{L}_n(x)) + e_n)$$

$$= (x, Tf(x)) = (x, f(x))$$

$$\Rightarrow (x, f(x)) \in \omega_n(\tilde{Q})$$

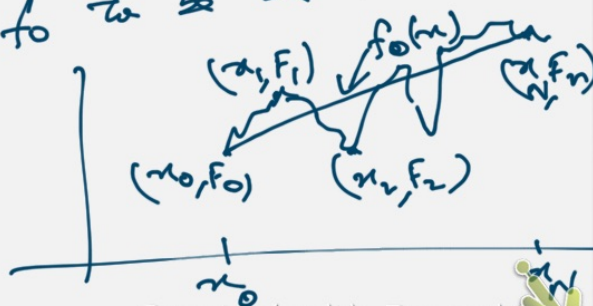
Hence, $\tilde{Q} \subseteq \bigcup_{n=1}^N \omega_n(\tilde{Q})$

$\therefore \tilde{Q} = \bigcup_{n=1}^N \omega_n(\tilde{Q})$

Since, $\tilde{Q}$ is a nonempty compact subset of $\mathbb{R}^2$,
 by Theorem 1, $\tilde{Q}$ is the unique attractor of
 the IFS $\{\mathbb{R}^2, \omega_1, \omega_2, \dots, \omega_N\}$.

Defn: This function f , which is the unique
 fixed point of T , is called a fractal
 interpolation function corresponding to
 the data $\{(x_i, F_i) : i=0, 1, 2, \dots, N\}$.

- If we start with any $f_0 \in \mathcal{F}$, the
 $\{T^n f_0\}_{n=0}^\infty$ converges to f .
 We can take f_0 to be the linear
 function.



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Fractal dimension of fractal interpolation fns.

Theorem: Let $N \geq 2$ and let $\{(x_n, f_n) \in \mathbb{R}^2 : n=0, 1, 2, \dots, N\}$ be a set of data.
 Let $\{\mathbb{R}^2; w_1, w_2, \dots, w_N\}$ be an IFS associated with the data, where

$$w_n \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_n & 0 \\ c_n & d_n \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_n \\ f_n \end{pmatrix}, \quad n=1, 2, \dots, N.$$

The vertical scaling factors d_n satisfy $0 \leq d_n < 1$, and a_n, c_n, e_n, f_n are given as before.
 Let A denote the attractor of the IFS, so that A is the graph of a fractal interpolation function f as before.

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If $\sum_{n=1}^N |d_n| > 1$, and the interpolation points are not collinear, then the box-counting dimension of A is the unique real no D satisfying $\sum_{n=1}^N |d_n| a_n^{D-1} = 1$.

If the interpolation points are collinear, then the dimension is one.

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